

S2: Advanced Statistical Methods for Data Intensive Science

Coursework Assignment

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1 Introduction

The Antikythera mechanism, discovered in 1901 near a Greek island, includes a fragmented calendar ring whose original total hole count remains unknown (Figure 1). Scholars have debated whether it represented a solar calendar (365 holes) or a lunar calendar (354 holes).

This project utilises data extracted from X-ray imaging of the ring [1] and applies Bayesian inference to estimate the total number of holes on the complete ring. Two likelihood models are implemented: one reproduces the anisotropic Gaussian model introduced by Woan et al. [2], while the other is a new isotropic Gaussian model proposed in this work. We compare their performance to determine which assumption is better supported by the observed data.

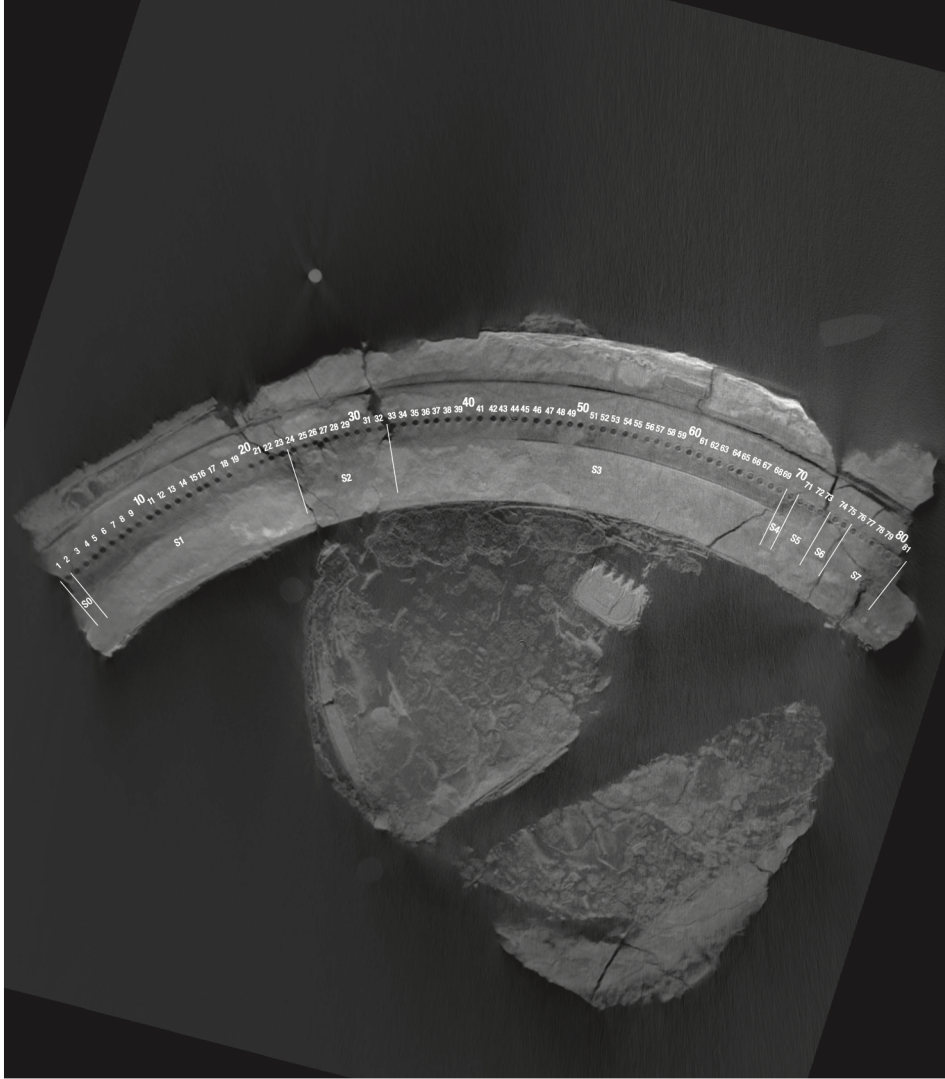


Figure 1: X-ray image of the Antikythera Mechanism's calendar ring, annotated by Budiselic et al. with holes and sections[1].

2 Data Visualisation

Figure 2 shows the measured positions of all 81 surviving holes on the calendar ring in the x - y plane, segmented into eight fractured sections. These coordinates constitute the observed data used in our Bayesian inference.

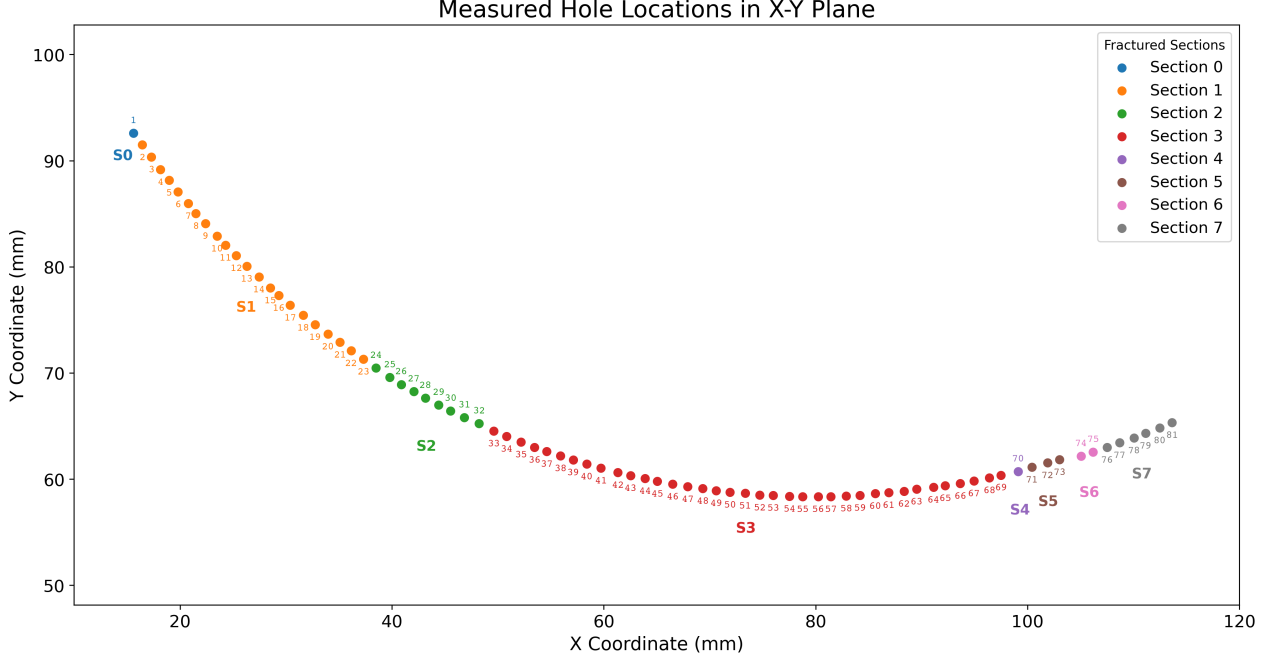


Figure 2: Measured (x, y) positions of the 81 visible holes in the Antikythera calendar ring, grouped into eight fractured sections (S0–S7) as defined by Budiselic et al. [1]. Each hole is numbered sequentially and coloured according to its section label.

3 Model Construction

Two likelihood-based error models are defined to characterise the hole locations, accounting for uncertainties introduced in both the hole construction and measurement processes. Specifically, we consider (a) Gaussian isotropic errors and (b) Gaussian anisotropic errors. This modelling approach is motivated by the Central Limit Theorem, as the total error likely arises from the accumulation of many small, independent perturbations. All relevant functions are implemented in `all_code/hole_model.py`.

Several general assumptions and notations are adopted in both models:

- Let N denote the total number of equally spaced holes along the calendar ring. Treating N as continuous accommodates the possibility of a local discontinuity in hole spacing due to mechanical constraints during fabrication.
- The calendar ring is modelled as a circle of radius r .
- Sections 0 and 4, which contain only a single hole each (see Figure 2), are excluded from model construction and inference, as single data points cannot constrain model parameters. Sections are indexed by $j \in \{1, 2, 3, 5, 6, 7\}$.
- The observed dataset is denoted as $\mathcal{D} = \{\vec{d}_{ij}\}$, which comprises 81 holes lying in the xy -plane and is indexed by $i \in \{1, \dots, 81\}$. Of these, 79 are considered informative, excluding hole 1 and hole 70.
- Due to mechanical fractures and possible misalignment, the eight segmented sections — although visually aligned in X-ray images — may exhibit translational and rotational displacement in the

xy -plane. Accordingly, the six effective sections are independently parameterised by $(\mathbf{x}_j, \mathbf{y}_j, \boldsymbol{\alpha}_j)$, where:

- $(\mathbf{x}_j, \mathbf{y}_j)$ represents the Cartesian coordinates of the local centre of the circular arc corresponding to section j ;
- $\boldsymbol{\alpha}_j$ denotes the theoretical angular offset of hole #1 (the hole located in Section 0), defined with respect to the global 0° direction, as observed from the local frame of section j .
- The theoretical angular position of hole i in section j is given by:

$$\phi_{ij} = \frac{2\pi(i-1)}{N} + \alpha_j,$$

where $\frac{2\pi}{N}$ represents the angular spacing between adjacent holes.

- The corresponding ideal hole location in Cartesian coordinates is:

$$\vec{\mathbf{r}}_{ij} = \begin{bmatrix} \mathbf{x}_j + r \cos(\phi_{ij}) \\ \mathbf{y}_j + r \sin(\phi_{ij}) \end{bmatrix}$$

- The measurement or construction error for hole i in section j is represented as:

$$\vec{\mathbf{e}}_{ij} = \vec{\mathbf{r}}_{ij} - \vec{\mathbf{d}}_{ij},$$

and all errors $\vec{\mathbf{e}}_{ij}$ are assumed to be independent across holes.

- The likelihood function in both models follows the general form of a multivariate Gaussian:

$$\mathcal{L}(\mathcal{D} \mid \theta) = \prod_{i,j} \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}\vec{\mathbf{e}}_{ij}^\top \Sigma^{-1} \vec{\mathbf{e}}_{ij}\right),$$

where $\vec{\mathbf{e}}_{ij}$ is the error vector for hole i in section j , and Σ is a 2×2 covariance matrix, defined differently for each model.

In total, both models share 20 common parameters: N , r , and $(\mathbf{x}_j, \mathbf{y}_j, \boldsymbol{\alpha}_j)$ for $j = 1, 2, 3, 5, 6, 7$. These parameters jointly define the functions ϕ_{ij} , $\vec{\mathbf{r}}_{ij}$, and $\vec{\mathbf{e}}_{ij}$, and are fitted to the common dataset $\mathcal{D} = \{\vec{\mathbf{d}}_{ij}\}$ with $i \in \{1, \dots, 81\}$, excluding $i = 1$ and $i = 40$.

a Gaussian Isotropic Model: `model_XY`

The isotropic model characterises the error vector of each hole location, $\vec{\mathbf{e}}_{ij} = (e_{x,ij}, e_{y,ij})$, as a 2D Gaussian distribution with zero mean and isotropic covariance:

$$\vec{\mathbf{e}}_{ij} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{\text{iso}}^2 & 0 \\ 0 & \sigma_{\text{iso}}^2 \end{bmatrix}\right)$$

where σ_{iso} denotes the standard deviation of error, assumed equal in all directions in the xy -plane.

We refer to this model as `model_XY`, as it assumes isotropic errors in the xy -plane, with the x and y directions chosen as the principal axes for analysis. The model includes 21 parameters in total:

$$\theta_{XY} = [N, r, \sigma_{\text{iso}}, \{x_j, y_j, \alpha_j\}_{j=1,2,3,5,6,7}].$$

- **Likelihood**

The likelihood of the observed data $\mathcal{D} = \{\vec{\mathbf{d}}_{ij}\}$ under the parameter set θ_{XY} is given by:

$$\mathcal{L}(\mathcal{D} \mid \theta_{XY}) = (2\pi\sigma_{\text{iso}}^2)^{-n} \prod_j \prod_{i \in j} \exp\left(-\frac{e_{x,ij}^2 + e_{y,ij}^2}{2\sigma_{\text{iso}}^2}\right), \quad (1)$$

where $n = 79$ is the number of effective hole observations.

- **Prior**

Due to limited prior knowledge of the calendar ring configuration, we adopt uniform priors over physically plausible ranges informed by existing observations (see Table 2):

$$\pi(\theta_{XY}) = \begin{cases} \frac{1}{b-a}, & \text{if } \theta_{XY} \in [a, b] \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

- **Posterior**

By Bayes' theorem, the posterior distribution over parameters is proportional to the product of the likelihood and prior:

$$P(\theta_{XY} \mid \mathcal{D}) \propto \mathcal{L}(\mathcal{D} \mid \theta_{XY}) \cdot \pi(\theta_{XY}),$$

where the normalisation constant $P(\mathcal{D})$ (the evidence) is omitted since it does not depend on the parameters.

b Gaussian Anisotropic Model: `model_RT`

The anisotropic model characterises the error vector of each hole location, $\vec{e}_{ij}$, as a bivariate Gaussian distribution whose principal axes are aligned with the radial and tangential directions at that hole on the circle [2]. The error distribution has the same mean as in `model_XY`, but a different covariance matrix:

$$\vec{e}_{ij} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_t^2 \end{bmatrix} \right),$$

where σ_r and σ_t denote the standard deviations in the radial and tangential directions, respectively. We refer to this model as `model_RT`, as it assumes independent anisotropic errors in the radial–tangential frame. This model contains 22 parameters:

$$\theta_{RT} = [N, r, \sigma_r, \sigma_t, \{x_j, y_j, \alpha_j\}_{j=1,2,3,5,6,7}].$$

- **Likelihood**

The likelihood of the observed data $\mathcal{D}$ under parameter set θ_{RT} is given by:

$$\mathcal{L}(\mathcal{D} \mid \theta_{RT}) = (2\pi\sigma_r\sigma_t)^{-n} \prod_j \prod_{i \in j} \exp \left(-\frac{e_{r,ij}^2}{2\sigma_r^2} - \frac{e_{t,ij}^2}{2\sigma_t^2} \right) \quad (3)$$

where $n = 79$ is the number of effective hole observations. The term $(e_{r,ij}, e_{t,ij})$ are the radial and tangential components of the error vector $\vec{e}_{ij}$, obtained by projection:

$$\begin{aligned} e_{r,ij} &= \vec{e}_{ij} \cdot \hat{r}_{ij}, \\ e_{t,ij} &= \vec{e}_{ij} \cdot \hat{t}_{ij}, \end{aligned}$$

and the unit vectors in the local radial–tangential coordinate frame are defined as:

$$\hat{r}_{ij} = \begin{bmatrix} \cos \phi_{ij} \\ \sin \phi_{ij} \end{bmatrix}, \quad \hat{t}_{ij} = \begin{bmatrix} \sin \phi_{ij} \\ -\cos \phi_{ij} \end{bmatrix}.$$

- **Prior**

As in `model_XY`, we adopt uniform priors (Equation 2) for all parameters, with bounds chosen based on existing observations (see Table 2).

- **Posterior**

Using Bayes' theorem, the posterior distribution over the parameters is:

$$P(\theta_{RT} \mid \mathcal{D}) \propto \mathcal{L}(\mathcal{D} \mid \theta_{RT}) \cdot \pi(\theta_{RT}),$$

where the proportionality ignores the marginal evidence $P(\mathcal{D})$.

Parameters	Uniform Prior Range [a, b]
N	[300, 400]
r	[0, 500]
x_j	[0, 200]
y_j	[0, 200]
α_j	$[-\pi, \pi]$
σ_{iso} (model_XY)	[0, 5]
σ_r (model_RT)	[0, 5]
σ_t (model_RT)	[0, 5]

Table 1: Uniform prior ranges used for both model.

4 Maximum Likelihood Estimation

a Calculate Gradient

In order to estimate the most likely parameter values, the maximum likelihood estimation problem is reformulated as a minimisation of the negative log-likelihood:

$$\arg \min_{\theta} -\log \mathcal{L}(\mathcal{D} \mid \theta).$$

Automatic differentiation is used to compute the gradients of the log-likelihood with respect to all model parameters. This technique systematically applies the chain rule of calculus at the level of elementary operations. In this work, gradients are computed using the `jax.grad` function from the JAX library [3].

To enable this, all functions discussed in Section 3 have been implemented in JAX-compatible form in `all_code/hole_model.py`. By constructing a computational graph of the log-likelihood function, JAX is able to compute exact analytical gradients via reverse-mode automatic differentiation, without requiring manual derivations or finite-difference approximations.

Code tests are conducted to validate the correctness of gradients computed by `jax.grad`, using the `check_grads` utility from `jax.test_util`. This function compares gradients obtained via automatic differentiation with those estimated using finite-difference methods, under both forward-mode and reverse-mode differentiation.

```
print("Testing gradients for model...")
check_grads(
    lambda params: NLL_jax(params, data),
    (params_test,),
    order=1,
    modes=["fwd", "rev"],
    atol=5e-3,
    rtol=5e-3)
print("Passed gradient check.")
```

The parameters `atol` and `rtol` define the absolute and relative tolerance thresholds for the comparison. A gradient component is considered accurate if the difference between the value computed by automatic differentiation and the finite-difference estimate satisfies the condition:

$$\text{difference} < \text{tolerance} = \text{atol} + \text{rtol} \times |\text{expected value}|, \quad (4)$$

where the “expected value” refers to the finite-difference approximation of the derivative.

The test passed for all parameters in both models under the specified tolerances (`atol = 5e-3`, `rtol = 5e-3`), confirming that the gradients computed via automatic differentiation are numerically consistent with the expected derivatives.

In addition, a test suite is provided in the `tests` directory to enable quick verification of gradient correctness. To execute the test suite, run the following command from the root directory of the project:

```
pytest -s tests/
```

The suite contains two tests, each corresponding to a gradient check for all parameters of one of the two models. Both tests should pass if the gradient computations via JAX are correctly implemented (see Figure 3).

```
(s2env) yilan@YilandeMacBook-Pro S2_coursework % pytest -s tests/
===== test session starts =====
platform darwin -- Python 3.9.6, pytest-8.3.5, pluggy-1.5.0
rootdir: /Users/yilan/Desktop/S2_coursework
plugins: anyio-4.9.0, typeguard-4.4.2
collected 2 items

tests/test_Qd.py ..

===== 2 passed in 2.41s =====
```

Figure 3: Output of running the test suite using `pytest`. Two gradient check tests were executed and successfully passed, verifying the correctness of the automatic differentiation implementation for both models.

b Maximise Likelihood

With gradients supplied via `jax.grad`, the parameters for both `model_XY` and `model_RT` were estimated by minimising the negative log-likelihood using the L-BFGS-B algorithm, as implemented in `scipy.optimize.minimize` [4]. L-BFGS-B is a quasi-Newton method that approximates second-order curvature information to accelerate convergence and supports box constraints, making it well suited for the medium-scale optimisation problem in this project.

To independently cross-validate the optimisation results, the `iminuit` [5] optimiser was also applied to the same model functions, using identical initial parameter values. The estimates obtained from `iminuit` were in general agreement with those from `scipy.optimize.minimize`, supporting the reliability of the maximum likelihood procedure (Table 2).

Parameter	model_XY (JAX+scipy)	model_XY (iminuit)	model_RT (JAX+scipy)	model_RT (iminuit)
N	354.2	355.0	355.2	355.2
r	77.2	77.3	77.4	77.3
σ_{iso}	0.10	0.10	N/A	N/A
σ_r	N/A	N/A	0.03	0.03
σ_t	N/A	N/A	0.12	0.12
x_1	79.5	79.7	79.7	79.7
x_2	82.1	79.9	80.0	79.9
x_3	79.8	79.9	79.9	78.9
x_5	77.6	81.5	78.4	81.4
x_6	78.1	81.0	78.2	81.5
x_7	77.6	83.2	82.6	83.2
y_1	136.0	136.1	136.1	136.0
y_2	134.2	135.7	135.7	135.7
y_3	135.5	135.7	135.8	135.7
y_5	134.8	136.1	135.3	136.1
y_6	134.4	135.8	134.7	135.8
y_7	133.8	136.4	136.2	136.4
α_1	-2.56	-2.56	-2.56	-2.56
α_2	-2.60	-2.56	-2.56	-2.56
α_3	-2.56	-2.56	-2.56	-2.56
α_5	-2.53	-2.58	-2.54	-2.58
α_6	-2.53	-2.57	-2.53	-2.57
α_7	-2.52	-2.60	-2.59	-2.60

Table 2: Maximum likelihood estimates of model parameters under two error models (XY and RT), using two optimisation methods (JAX+Scipy and iminuit). Length units are millimetres and angle units are radians.

Figure 4 presents the predicted hole positions obtained using JAX and SciPy under the maximum likelihood estimates for both the isotropic (`model_XY`) and anisotropic (`model_RT`) models, overlaid on the observed data. While both models show good overall agreement with the measurements, small section-specific discrepancies—particularly in Section 7—are evident in the predictions from `model_XY`. This highlights the limited flexibility of the isotropic model in capturing the variations in the data.

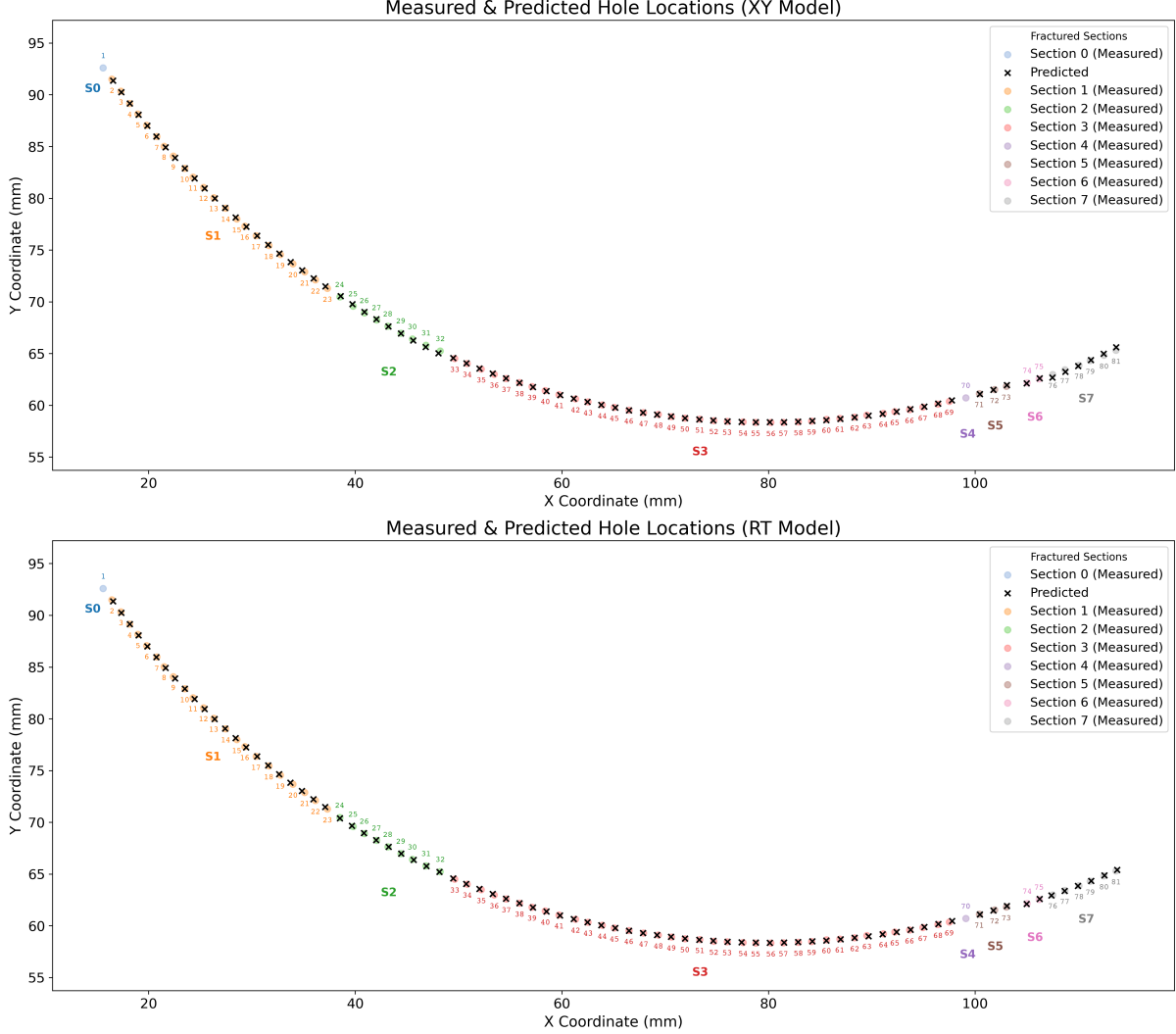


Figure 4: Measured and maximum likelihood predicted hole positions under the two error models. Measured data points are colour-coded by section number. Predicted locations (black crosses) correspond to the maximum likelihood estimates computed using the JAX + `scipy.optimize.minimize` implementation for each model. Note that holes 1 and 70 are omitted from the prediction as they were not included in the model fitting.

From Table 2, we observe that the maximum likelihood estimate of N is approximately 355 for the given measurements, which corresponds to the lunar calendar. However, without a valid uncertainty range or a quantified probability for alternative values $N = 365$, we cannot conclusively rule out the solar calendar hypothesis. To address this, posterior sampling was performed in the following sections to quantify the uncertainty and assess the relative plausibility of different values of N .

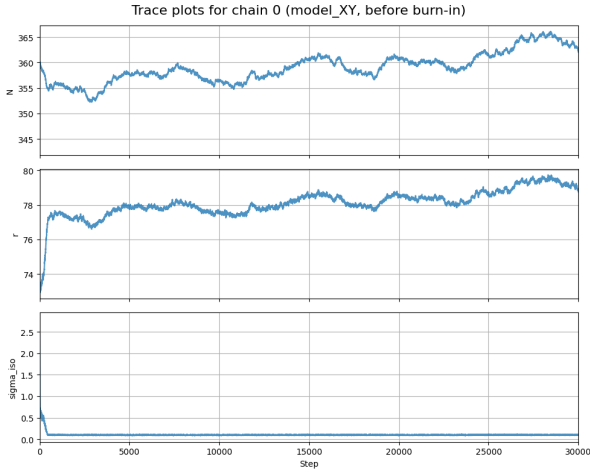
5 Posterior Sampling

a Hamiltonian Monte Carlo Posterior Sampling

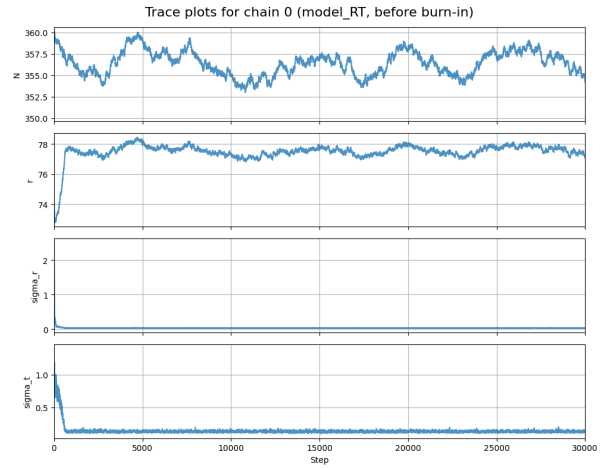
Hamiltonian Monte Carlo (HMC) was employed to sample the posterior distributions for both `model_XY` and `model_RT`. The implementation used `blackjax.hmc`, a performant and flexible JAX-based HMC sampler.

Both posterior models were implemented as described in Section 3. Several combinations of step size and number of integration steps were explored to optimise sampling performance. The final configuration used a step size of 0.0003 and 200 integration steps, which yielded acceptance rates of 78.2% for `model_XY` and 90.1% for `model_RT`, both well within the desirable range.

For each model, HMC was run using 30 parallel chains (walkers), each generating 200,000 samples. The first 5,000 samples of each chain were discarded as burn-in (see Figure 5). Long chains were used to ensure sampling quality, as the autocorrelation time was observed to be high when shorter chains were employed. Under the chosen configuration, the maximum autocorrelation times were 1.07 and 1.01 for `model_XY` and `model_RT`, respectively, indicating that most samples can be considered effectively independent.



(a) First chain for `model_XY`.



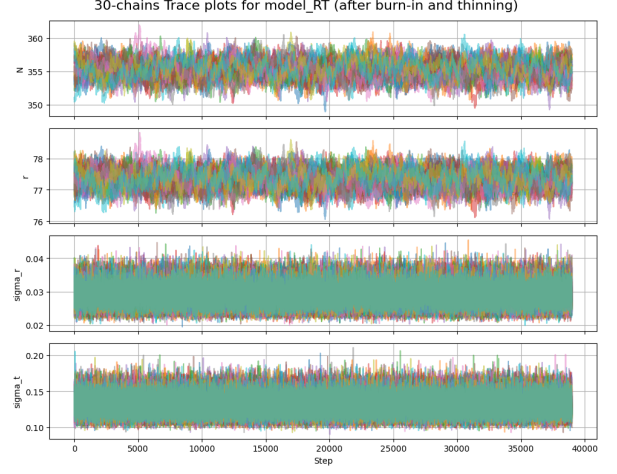
(b) First chain for `model_RT`.

Figure 5: Trace plots of the first 30,000 samples from the first chain for selected parameters, used to assess convergence and determine appropriate burn-in lengths for both models.

To ensure decorrelation and avoid over-crowding in visualisations, all chains were conservatively thinned by a factor of 5. Figure 6 shows trace plots of the key parameters (N , r , and σ) after thinning. Additionally, Gelman–Rubin diagnostics were computed to assess convergence, with $\hat{R}$ values below 1.2 for nearly all parameters in both models.



(a) Trace plots for `model_XY`.



(b) Trace plots for `model_RT`.

Figure 6: Trace plots of the thinned chains for the parameters N , r , and σ in both models. The plots show well-mixed samples with no apparent drift, supporting the convergence diagnostics based on the Gelman–Rubin statistic.

These results together indicate good mixing and convergence of the HMC chains. After thinning and burn-in removal, 1,170,000 posterior samples remained for each model. Figures 7 and 8 display the corresponding corner plots for the intrinsic parameters (N , r , and σ). Table 3 presents the posterior means and standard deviations for all model parameters, while Table 4 summarises the credible intervals for the intrinsic parameters N , r , and σ .

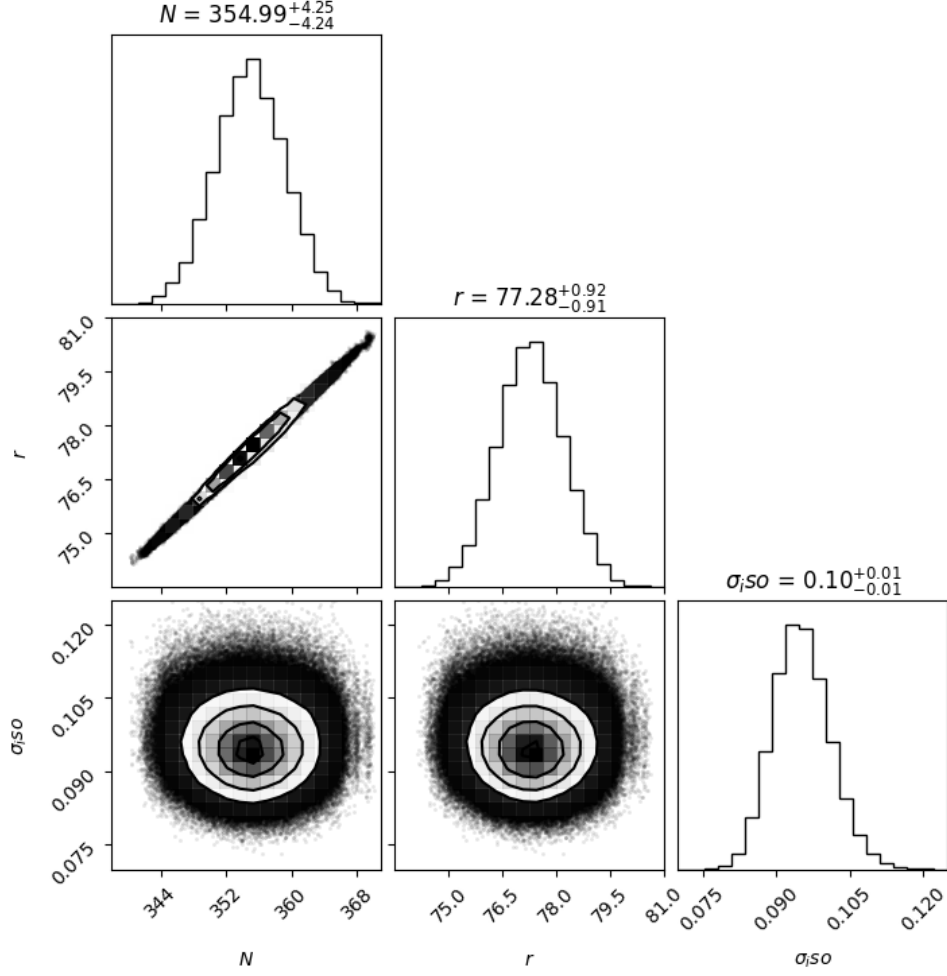


Figure 7: Corner plot of the posterior distributions for `model_XY` (isotropic). The plot includes the marginal and joint distributions of N , r , and σ_{iso} , sampled via HMC. Median values and 68% credible intervals are annotated. The inferred correlation between N and r is evident, while the isotropic error model captures only a single uncertainty scale across all directions. Length units are millimetres.

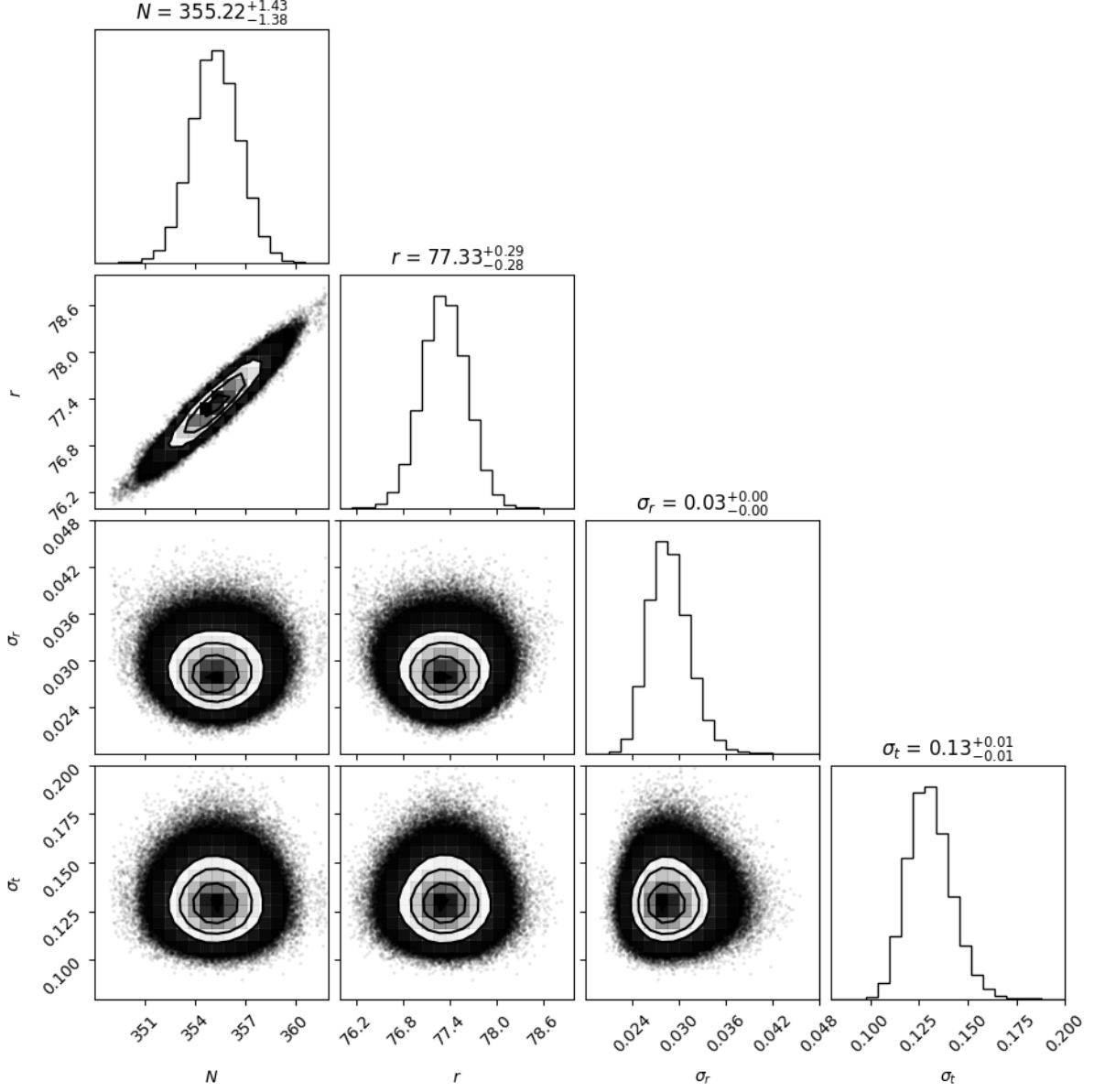


Figure 8: Corner plot of the posterior distributions for `model_RT` (anisotropic). The plot shows marginal and joint distributions of the intrinsic parameters N , r , σ_r , and σ_t , sampled via HMC. Median values and 68% credible intervals are annotated. The inferred correlation between N and r is evident. The anisotropic structure allows separate modelling of radial and tangential uncertainties. Length units are millimetres.

Table 3: Posterior means and standard deviations for all parameters in `model_XY` and `model_RT`. Length units are millimetres. Angular units are radians.

Parameter	model_XY	model_RT
N	354.99 ± 4.19	355.24 ± 1.41
r	77.29 ± 0.90	77.34 ± 0.29
σ_{iso}	0.10 ± 0.01	N/A
σ_r	N/A	0.03 ± 0.00
σ_t	N/A	0.13 ± 0.01
x_1	79.64 ± 0.65	79.69 ± 0.20
x_2	79.88 ± 0.74	79.91 ± 0.23
x_3	79.86 ± 0.11	79.86 ± 0.04
x_5	81.18 ± 3.84	81.45 ± 1.13
x_6	81.13 ± 7.40	81.51 ± 2.54
x_7	83.24 ± 1.28	83.22 ± 0.39
y_1	136.02 ± 0.67	136.03 ± 0.21
y_2	135.67 ± 0.87	135.71 ± 0.27
y_3	135.65 ± 0.92	135.71 ± 0.29
y_5	135.87 ± 1.38	136.10 ± 0.41
y_6	135.25 ± 2.58	135.79 ± 0.87
y_7	136.36 ± 0.97	136.42 ± 0.31
α_1	-2.56 ± 0.00	-2.56 ± 0.00
α_2	-2.56 ± 0.01	-2.56 ± 0.00
α_3	-2.56 ± 0.01	-2.56 ± 0.00
α_5	-2.58 ± 0.05	-2.58 ± 0.02
α_6	-2.57 ± 0.10	-2.57 ± 0.04
α_7	-2.60 ± 0.02	-2.60 ± 0.01

Table 4: Posterior 68% and 94% credible intervals for intrinsic parameters, reported with lower and upper bounds. Length units are millimetres. Angular units are radians.

Parameter	model_XY 68% CI		model_XY 94% CI		model_RT 68% CI		model_RT 94% CI	
	Lower	Upper	Lower	Upper	Lower	Upper	Lower	Upper
N	350.75	359.24	347.24	362.97	353.84	356.65	352.59	357.90
r	76.37	78.20	75.62	79.01	77.05	77.62	76.80	77.88
σ_{iso}	0.09	0.11	0.08	0.11	N/A	N/A	N/A	N/A
σ_r	N/A	N/A	N/A	N/A	0.03	0.03	0.02	0.03
σ_t	N/A	N/A	N/A	N/A	0.12	0.14	0.11	0.15

b Posterior Predictive Distribution

Posterior predictive distributions for hole 41 in Section 3 were visualised for both the isotropic and anisotropic models, as shown in Figure 9. These predictions were generated by substituting each set of HMC-sampled parameters into the respective model and drawing samples from the corresponding Gaussian distributions. In both cases, the observed hole 41 position lies well within the 68% posterior predictive region, indicating good model accuracy.

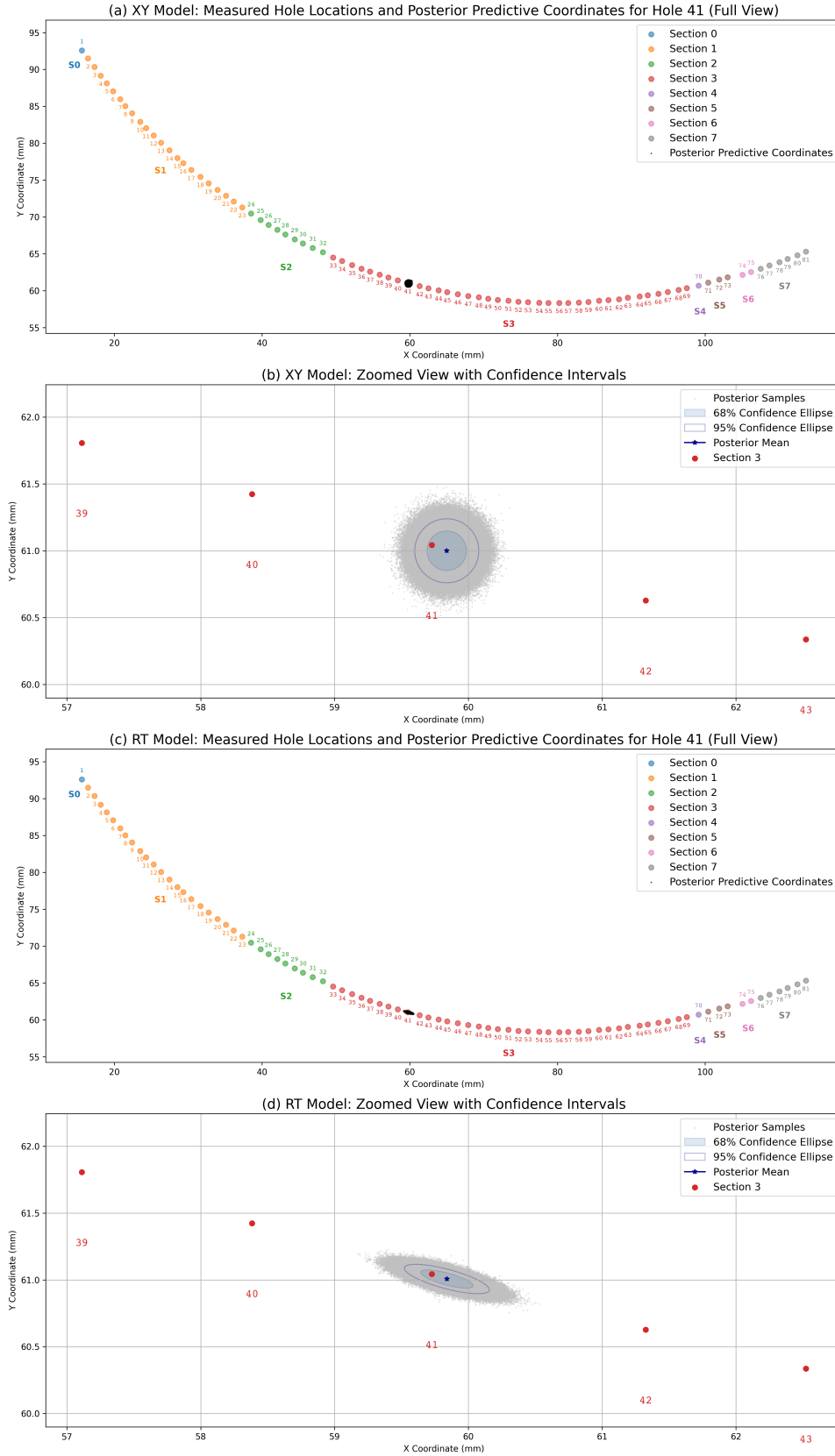


Figure 9: Posterior predictive visualisation of hole 41 using the isotropic (`model_XY`) and anisotropic (`model_RT`) models. (a) and (c) show full views of all measured hole locations, with posterior predictive samples for hole 41 overlaid in black. Sections are colour-coded by index. (b) and (d) show zoomed-in views of hole 41, including posterior samples (grey), posterior means (blue), and confidence ellipses (68% in light, 95% in dark). The shape of the predictive distributions reflects the isotropic or anisotropic error assumptions of each model.

c Hole Number Inference

The posterior sampling results of `model_RT`, as shown in Table 3, closely replicate the findings of the reference study [2], thereby validating the correctness and reproducibility of the posterior sampling implementation. A strong positive correlation between N and r is observed, as expected: if the angular spacing is fixed, a greater number of holes necessarily implies a larger ring radius.

Although the posterior means of parameters estimated by `model_XY` and `model_RT` are broadly similar, the credible intervals inferred from `model_XY` are significantly wider—approximately three times larger on average than those from `model_RT`. This suggests that the isotropic model provides substantially weaker constraints on parameter estimates.

Crucially, when treating the total hole count N as a continuous parameter, the value $N = 365$ does not lie within the 94% credible intervals in either model (see Table 4; `model_XY`: [347.24, 362.97], `model_RT`: [352.59, 357.90]). This exclusion provides strong Bayesian evidence against the hypothesis that the calendar ring was designed with 365 holes. Instead, both models strongly favour values near $N = 354$, consistent with the lunar calendar hypothesis. The credible intervals obtained through rigorous Bayesian inference thus offer quantitative support for the lunar design interpretation of the Antikythera mechanism.

6 Model Comparison

a Covariance Matrix

In this project, the 2D covariance matrix Σ plays a key role in modelling the distribution of measurement errors in the spatial coordinates of the calendar ring holes. It captures the variance along the defined principal axes (through the diagonal elements of Σ) and reflects how much uncertainty is allowed in each direction. When the diagonal terms are equal, the model assumes isotropic noise, which means the error is equally tolerated in all directions (as in `model_XY`); when they differ, it becomes anisotropic, allowing direction-dependent flexibility in capturing the error structure (as in `model_RT`). In both cases, the off-diagonal terms are set to zero. This implies that errors along the principal directions are assumed independent.

Incorporating directional information allows for a more precise probabilistic treatment of measurement uncertainty. This, in turn, improves the model’s adaptability to different geometrical conditions, particularly when anisotropic noise is present.

b *A Priori* Model Choice

In this project, the radial/tangential anisotropic model (`model_RT`) is initially considered a suitable choice, based on the following considerations.

- **Realistic Consideration**

The errors in our model primarily arise from two sources: measurement uncertainty and imperfections during the original manufacturing of the calendar ring. Given that the holes lie on a circular path, it is reasonable to suppose that the construction began from a fixed central point, with holes drilled by rotating a tool at a constant radius. Under this assumption, radial errors are expected to be naturally small, since both the centre and the radius were likely fixed during construction. In contrast, tangential errors depend on the precise control of the rotation angle, which is more susceptible to inaccuracies—especially considering the limited tooling precision available in antiquity. Therefore, it is a realistic choice to model the error using an anisotropic covariance matrix aligned with the radial and tangential directions.

This assumption is further supported by the observed results, where σ_t is significantly larger than σ_r (Table 3).

- **Statistical Consideration**

The anisotropic assumption carries more information and provides greater flexibility for the model to fit the data in parameter space, hence more expressive. When the true error properties are unknown, it is preferable to allow such flexibility, as the isotropic model (`model_XY`) is nested within the anisotropic one (`model_RT`). If the errors are truly isotropic, the model will naturally yield $\sigma_r = \sigma_t$, effectively reducing to the isotropic case. The reverse, however, is not true. Therefore, adopting the anisotropic model introduces no risk but offers the potential for improved modelling accuracy.

c Data-Driven Model Selection

c.1 Insights from Posterior Sampling

Posterior sampling results show that in the anisotropic radial–tangential model, the tangential error is approximately three times larger than the radial error. This confirms the hypothesised asymmetry in construction precision and provides strong evidence in favour of `model_RT`. If the underlying error were truly isotropic, the radial and tangential components would be expected to exhibit comparable magnitudes.

Although both models achieve high absolute precision relative to the spacing between holes, `model_XY` exhibits considerably higher posterior variance. This reflects its inability to capture direction-dependent errors. In contrast, the radial–tangential decomposition in `model_RT` allows for more structured error modelling and yields more constrained and physically interpretable inferences.

Figure 9 visualises the posterior predictive distributions for hole 41 under both models. While `model_XY` produces a circular uncertainty region consistent with isotropic noise, `model_RT` yields an elongated ellipse that aligns with the dominant tangential deviations observed in the data. This visual distinction further supports the anisotropic error assumptions underlying `model_RT`.

c.2 Savage-Dickey Density Ratio

In this study, `model_XY` can be regarded as a nested version of `model_RT`. To demonstrate this, we introduce a new variable $\epsilon = \sigma_t - \sigma_r$. When $\epsilon = 0$ (i.e., $\sigma_r = \sigma_t$), the anisotropic model reduces exactly to the isotropic Gaussian model defined by `model_XY`.

Given that both models share the same priors on the overlapping parameters, the Savage–Dickey density ratio provides a straightforward method for computing the Bayes factor between the two nested models. This ratio takes the form:

$$\mathcal{B}_{\text{XY,RT}} = \frac{P(\epsilon = 0 \mid \mathcal{D}, \text{model_RT})}{P(\epsilon = 0 \mid \text{model_RT})},$$

where the numerator is the posterior density at $\epsilon = 0$, and the denominator is the corresponding prior density.

This ratio quantifies the relative evidence in favour of the simpler model (`model_XY`) compared to the more complex one (`model_RT`). A small value (ratio < 1) of $\mathcal{B}_{\text{XY,RT}}$ indicates support for the more flexible anisotropic model.

To estimate this ratio, HMC posterior samples obtained in Section 5(a) were used to construct a kernel density estimate (KDE) of the posterior distribution over ϵ . The resulting estimate gives $P(\epsilon = 0 \mid \mathcal{D}, \text{model_RT}) \approx 0.0$, with the KDE visualisation shown in Figure 10. Hence,

$$\mathcal{B}_{\text{XY,RT}} = 0.$$

This result provides overwhelming evidence in favour of the anisotropic radial–tangential error model over the isotropic alternative, demonstrating that the additional flexibility of `model_RT` is not only justifiable, but also strongly supported by the data.

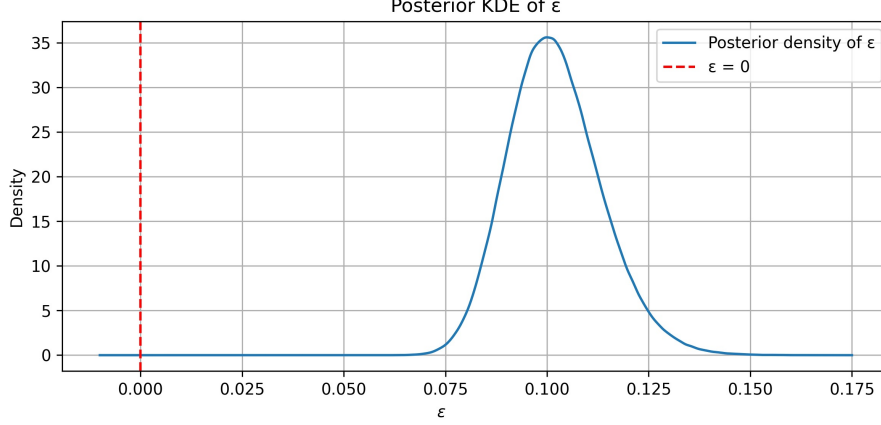


Figure 10: Kernel density estimate (KDE) of the posterior distribution of $\epsilon = \sigma_t - \sigma_r$ under `model_RT`. The red dashed line indicates $\epsilon = 0$, which corresponds to the special case of isotropic errors (i.e., $\sigma_t = \sigma_r$). The posterior density at $\epsilon = 0$ is effectively zero, demonstrating that the isotropic model is not supported by the posterior samples. This result enables the use of the Savage–Dickey density ratio, yielding a Bayes factor $\mathcal{B}_{\text{XY,RT}} = 0$, indicating decisive preference for the anisotropic model.

7 Summary

This project investigates the spatial distribution of holes on the calendar ring of the Antikythera Mechanism in order to evaluate competing calendar hypotheses. By modelling the hole positions in two dimensions under a Gaussian error framework, we find that an anisotropic covariance structure provides a better fit to the data. The analysis supports a 354-hole configuration, consistent with a lunar calendar system. These findings offer valuable insight into the calendar systems and technological capabilities of ancient societies.

8 Declaration

During the development of this project, I made use of Github Copilot’s autocompletion feature to assist in coding and generating docstrings for functions. Additionally, ChatGPT was used to support debugging. For the final report, ChatGPT helped in generating LaTeX code in desired formats.

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